

Supplementary material for “Spacetime quantities from causal order: an operational ladder with preregistered validations at finite density”

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This document carries the article’s conventions table (section S1) and the execution-provenance record of the preregistered stages (section S2). Section numbers without an S prefix refer to the article.

S1. Conventions and normalizations

Table S1. Conventions and normalizations fixed in the foundation modules.

Quantity	Convention used
Causal relation	null-inclusive J^+ : i precedes j iff $t_j - t_i > 0$ and $(dt)^2 - (dx)^2 \geq -\text{atol}$, $\text{atol} = 1 \times 10^{-12}$; strict in time (irreflexive)
Longest chain	endpoint-exclusive count (interior elements, two fewer than the count including both endpoints); (1+1)D normalization $L \sim \sqrt{2\rho}\tau$ (asymptotic normalized value $\sqrt{2} \approx 1.414$), finite-size corrected
Interval volume, (1+1)D	open Alexandrov count K ; $V = K/\rho$; $V = \tau^2/2$, so $\tau_{\text{est}} = \sqrt{2K/\rho}$
Myrheim–Meyer	ordering fraction $r = (\text{related ordered pairs})/(N(N-1))$ inverted against $f(d) = \Gamma(d+1)\Gamma(d/2)/(4\Gamma(3d/2))$; equals the standard unordered form up to a factor 2 on both sides
Radar coordinates	τ_- = latest preceding tick, τ_+ = earliest succeeding tick; radar time $= (\tau_+ + \tau_-)/2$, radar distance $= (\tau_+ - \tau_-)/2$
Density (finite- N)	fixed- N sprinkle with empirical density; Poisson and fixed- N binomial sampling models both reported

The conventions in table S1 are stated so the reported constants (e.g. the $\sqrt{2}$ chain normalization, the interval-volume factor) are unambiguous; each is fixed in the foundation modules.

S2. Capstone execution provenance

The section 6 result is reproducible not merely in the sense that its code is committed, but in the sense that the *order of operations*—freeze before execution, execution before recording—is provable from the repository’s commit graph and enforced by mechanical gates.

Commit chain. Design and probe chain: PR #38–#48. Preregistration: preparation commit (PR #49), S1 cost measurement (PR #50), then $P' = 51875a2$ (a

one-source seed-ledger fix) $\rightarrow$ preflight executed on a clean checkout of P' , its artifact recording `code_version` = P' and the SHA-256 digests of every execution-relevant source file $\rightarrow$ final-freeze commit F = b858b08 carrying the certification artifact and the freeze manifest (preflight digest, `preflight_code_version` = P' , the S1 price sentence, the frozen sizes and seeds; PR #51) $\rightarrow$ the campaign executed on a checkout of exactly F (not a descendant merge commit, so the recorded executing HEAD is F itself) $\rightarrow$ results commit R = 5126ddf, a direct child of F (PR #52). A test asserts $P' \prec F \prec R$ with `git merge-base --is-ancestor`.

Gates. Stochastic modes refuse a dirty worktree at entry. The campaign mode refuses to run unless (i) the freeze manifest exists, (ii) its recorded preflight digest matches the committed preflight artifact, and (iii) the source digests certified at preflight entry match the checked-out files—captured at entry and re-checked at exit, so a mid-run edit aborts the run with nothing recorded. Interrupted runs restart the same seed; an ambiguity violation blocks the stage rather than permitting a seed swap.

Artifacts. The complete section 6 evidence bundle, with digests, is enumerated in `artifact_manifest.md`; the per-sentence mapping from manuscript text to evidence file is recorded there as the citation-to-artifact inventory.

Frozen sentences. Each preregistered stage fixes, before execution, the sentence its verdict would license, and the campaign records that sentence in its results artifact. Those originals are written in Korean, the working language of this program's records; every frozen sentence printed in section 6.4 and section 6.7 is therefore a translation, and the artifact holds the authoritative text. What keeps the two from drifting apart is not editorial care: a contract test reads each original from its artifact, holds it against the rendering the manuscript prints in its place, and fails if either side moves—so a translation cannot quietly soften, strengthen, or outrun the sentence the campaign actually licensed. The paper prints no Korean, and a separate test enforces that, because a script the reader cannot check would put the load-bearing wording beyond the reader's reach.

Section 6.7 (Schwarzschild extension) provenance. Freeze commit `ceed85d` (PR #57, merge `5cffd90`) carries the frozen rule, the fresh seed registration, the runner, the contract tests, and the 8-file content-addressed freeze manifest—and no results. The campaign executed once on a clean checkout of exactly `ceed85d`; the runner captured the git state at entry, refused a dirty tree, verified every manifest digest at entry, and re-verified both at exit before writing, so the artifact's recorded lineage is `start == end == ceed85d`, `clean`, `run_kind = fresh_observation`. The results commit (PR #58, merge `492e61f`) added the artifact, moved the campaign seed from fresh to observed in the ledger, and flipped the runner's campaign path to a deterministic replay whose output is owned by a separate file and can never replace the fresh observation. Because that flip legitimately changed two protocol-surface files, the *current* tree's manifest hashes the post-result replay surface; the manifest that governed the executed run is preserved byte-for-byte as `docs/prereg/p14_s4_executed_freeze_manifest.json`, and a contract test verifies that snapshot against the historical blobs at `ceed85d` directly. Interrupted runs restart the same seed; undecided pairs are absorbed by identified intervals rather than a stop rule.

Section 6.7 second stage (S5) provenance. Freeze commit `1c26224` (PR #60, merge `3aeb8e5`) carried the rule, the fresh seed registrations, the runner, and the power

certification; an execution review then caught an exact-versus-approximate chi-square mismatch in the certified resampling source, and the corrected freeze is `86e3674` (PR #61, merge `f286508`)—certification unchanged under the exact factor. The campaign executed once on a clean checkout of exactly `86e3674` (entry == exit git state, the 8-file content-addressed manifest verified at both ends, fail-closed CLI, `run_kind = fresh_observation`, curved seed 40000251 / flat seed 40000261). The results commit (PR #62, merge `d1e0d20`) added the artifact, moved both seeds to observed, flipped the campaign path to a replay whose output is owned by a separate file, and preserved the manifest that governed the executed run byte-for-byte as `docs/prereg/p14_s5_executed_freeze_manifest.json`, verified against the historical blobs at `86e3674` by a contract test; the *current* `p14_s5_freeze_manifest.json` hashes the post-result replay surface.

Auxiliary O4b instrument-audit provenance (oracle arc). The certified oracle chain is $O1 \rightarrow O2 \rightarrow O3$ (PR #64–#68): flight-time certification, certified cell integrator, then the O3 freeze (`785148e`) and its one approved execution producing the certified volume (`docs/prereg/p14_o3_volume.json`, executed surface `p14_o3_executed_freeze_manifest.json`). The first audit stage, O4, aborted at its G3 with no verdict (PR #69–#70); its replay diagnostic and G3 redesign reopened the preregistration (PR #71–#74), and O4b was frozen with new seeds (PR #75). The approved execution SHA is the freeze branch head `715865a`—not the merge commit. The campaign executed once from that clean exact checkout: G3a passed (71 wrapper-contract cases, 6,537 metered calls), `refs/o4b/reservation` was claimed (`46acee34`, retained permanently, spending seeds 40000401/40000411), G1 ran to 26,200,000 points and G2 to 1,072,696 points, and the run stopped at the pre-publication provenance re-read on a wiring defect—the claim callback returned `None`—publishing nothing. The preservation commit (PR #76, merge `83e366c`) committed the incident and checkpoint verbatim (`p14_o4b_incident.json`, `p14_o4b_checkpoint.json`), retired both seeds in the ledger, and snapshotted the executed surface byte-for-byte (`p14_o4b_executed_freeze_manifest.json`, byte-identical to the freeze blob at `715865a`). The recovery commit (PR #77, merge `50586dd`) published `p14_o4b_results.json` with `run_kind = recovered_completed_campaign` by re-applying the frozen decision functions to the preserved statistics—bit-identical to the preserved verdicts, inputs authenticated by blob SHA-256 pins and an incident/checkpoint cross-check, the remote reservation re-verified immediately before publication; per the frozen stage-verdict table (G1 and G2 both concordant, G3a valid, frozen n reached) the stage verdict is CONCORDANT. The maintenance commit (PR #78, merge `a04e860`) fixed the wiring defects—the claim return value, the budget stage label, an explicit incident `failure_point`, a named checkpoint RNG stream—with end-to-end regression tests over the real `main()` path, leaving the result, incident, checkpoint, and executed snapshot untouched. No campaign rerun and no same-seed re-entry occurred anywhere in this chain; the historical no-verdict records of the O4 abort and the O4b incident retain their original verdicts, with the recovery linked rather than retroactively reggraded.

Section 6.8 (mass ladder) provenance. Each rung executed once from a clean exact checkout of its own freeze head, with the executing commit recorded in the artifact at entry and at exit (`code.start == code.end`, clean tree): `c6eb85e` for $\mu = 0.1333$, `1575a8e` for $\mu = 0.1867$, `c421dce` for $\mu = 0.2400$, and `ed353d3` for $\mu = 0.4000$. The

results commits `6d3745c`, `cdfd6d9`, `6bcf4e6`, and `ce0030d` added each rung's artifact, and each freeze head is an ancestor of its results commit—the freeze-before-execution ordering is checked by `tests/test_paper_a_count_integration.py` with `git merge-base --is-ancestor`, per rung, exactly as for the capstone.